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HY USMLE REVIEW
PART IV



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HY USMLE Review – Part IV

- 72M + drooping of left side of face + decreased muscle strength and increased tendon reflexes of both left upper and lower extremities + Babinski on left + intact sensation; what's the most likely explanation for this patient's condition? → answer = lacunar infarct of internal capsule; produces "pure motor" stroke in most cases of posterior capsule infarct; thalamic stroke produces "pure sensory" effects; lacunar infarcts are usually due to Hx of HTN causes lipohyalinosis of lenticulostriate arteries.
- 24M + experiment glucose injected IV in two different scenarios: 1) when food is simultaneously consumed orally, and 2) when no food is simultaneously consumed; experiment shows that glucose is cleared faster from circulation when food is concurrently consumed orally; Q asks, which hormone is responsible for these findings? → answer = glucose-dependent insulintropic peptide (GIP); secreted in response to oral consumption of macronutrients (i.e., carbs, fats, protein); causes insulin secretion; in other words, insulin goes up more when food is consumed orally vs merely received parenterally → faster clearance of plasma glucose.
- 24M + receives 50g of glucose IV vs 50g of glucose orally; insulin goes up more when glucose consumed orally; why? → answer = glucose-dependent insulintropic peptide → greater insulin secretion.
- 6F + finds open bottle of aspirin; which of the following findings in the patient best suggests the dose consumed was toxic? → answer = increased respiratory rate; aspirin causes respiratory alkalosis acutely due to upregulation of respiratory centers (high pH, low CO₂, no change bicarb [too acute to change]); after roughly 20 minutes, a mixed metabolic acidosis-respiratory alkalosis is seen (low pH, low CO₂; low bicarb); the low bicarb **is not** compensation for the acute respiratory alkalosis; it is merely convenient that the salicylic acid itself is an acid and drops the bicarb; tachypnea is seen acutely as first sign (observed by healthcare practitioner); tinnitus can be reported as first symptom (patient-reported finding); Tx with sodium bicarb → increased excretion of aspirin via urinary alkalization.
- 30M + shooting groin pain + increased blood pressure + mother had surgery for thyroid cancer; what's most likely to establish the cause of the patient's HTN? → answer = urinary metanephrines; diagnosis is MEN 2A → pheochromocytoma + medullary thyroid carcinoma + primary

hyperparathyroidism; Tx pheo with phenoxybenzamine (irreversible alpha blocker); shooting groin pain is urolithiasis secondary to hypercalcemia; medullary thyroid cancer causes increased serum calcitonin. For quick comparison, MEN I → pancreatic tumor, pituitary tumor, primary hyperparathyroidism; MEN 2B → pheochromocytoma + medullary thyroid carcinoma + mucosal neuromas + Marfanoid body habitus.

- Neonate + strange hereditary disease + mutation analysis shows homozygous gene deletion; what's the most likely inheritance pattern? → answer = autosomal recessive; some students get caught up by saying, "Wait, we're supposed to know the inheritance pattern of this weird disease?" No. It's the fact that they say *homozygous* deletion, without mentioning that either parent has pathology, that implies the condition is autosomal recessive.
- Patient takes nedocromil for seasonal allergies; what's the MOA of this drug? → answer = blocks release of autacoids from cellular storage; student says wtf?; mast cell stabilizers like nedocromil and cromolyn sodium prevent the release of histamine, which acts as an autacoid, which means it's a biologic molecule that acts as a local hormone (i.e., acts near site of synthesis).
- 84F + unilateral hydronephrosis caused by ureteral compression by a vascular structure; what's the structure → answer = ipsilateral common iliac artery; can compress ureter leading to hydronephrosis; if the Q tells you the patient has cancer causing unilateral hydronephrosis, choose ovarian cancer impinging on ureter; if Q tells you patient has bilateral hydronephrosis due to cancer, choose cervical cancer (usually younger patient).
- 54M + progressive shortness of breath for four months + ascites + edema of lower extremities + bilateral pleural effusions + ejection fraction of 55%; what's the mechanism for this patient's increased fluid in the lungs? → answer = decreased plasma colloid osmotic pressure (decreased plasma oncotic pressure); this refers to either renal (nephrotic syndrome) or hepatic disease (decreased production of albumin); wrong answer is "increased capillary hydrostatic pressure" because ejection fraction is normal (55-70); this is normally the answer for pulmonary edema from left heart failure; "increased vascular permeability" would be the correct answer in ARDS (bilateral pulmonary infiltrates), infection (i.e., pneumonia +/- sepsis), or pulmonary embolism (local inflammation).

- Neonate + persistent cyanosis + normal Hct + normal RBC morphology + CXR and echo show no abnormalities; diagnosis? → answer = methemoglobinemia due to deficiency of cytochrome reductase B5; methemoglobinemia can lead to cyanosis; USMLE asks for cytochrome reductase B5 deficiency as cause of congenital methemoglobinemia; can lead to persistent cyanosis; should be noted that nitric oxide used for pulmonary hypertension in neonates can cause methemoglobinemia (oxidizes ferrous iron to ferric iron); acrocyanosis is normal blueish tinge of extremities at birth in some neonates (treatment is place under warming lights and do tactile stimulation); Tx for methemoglobinemia in most cases is methylene blue + vitamin C.
- 40F + fibromuscular dysplasia + GFR is within normal range; what explains this latter finding? → answer = increased efferent arteriolar resistance; RAAS will upregulate in response to decreased renal blood flow → angiotensin II constricts efferent arterioles, causing increased filtration fraction, where GFR is held constant despite lower blood flow to kidney.
- 19F + external compression of left renal artery but has normal renal perfusion pressure; what explains this latter finding? → answer = decreased *afferent* arteriolar resistance; just happens to be the answer they give in this hypothetical scenario; if the afferent perfusion is decreased due to compression of the renal artery, the only way perfusion pressure can still be unchanged is if the smaller afferent arterioles distal to the renal artery dilate; slightly unusual Q considering the normal physiologic response to reduced renal perfusion is simply efferent arteriolar constriction via increased RAAS activation; but in this latter scenario, the result is maintenance of GFR *despite* reduced renal perfusion, rather than a maintenance of renal perfusion.
- 27M + from Pennsylvania + Hx of splenectomy + RBCs show intraerythrocytic rings; diagnosis? → answer = Babesia; causes Malaria-like hemolytic disorder in patients who've never left the United States; ring forms and Maltese crosses can be seen within RBCs; if the Q tells you the patient went to Africa, choose malaria, not Babesia. But if the patient didn't leave the US, malaria is the **wrong** answer.
- How many kilocalories in one gram of fat, carbs, protein, and ethanol? → answer = 9 kcal/g of fat; 4 for both carbs and protein; 7 for ethanol; USMLE Q will give you some nutrition Q where you have to do math, and it will rely on you knowing this nutritional info to get the Q right.

- Q asks for the mechanism via which insulin acts; answer = tyrosine kinase (MAP); USMLE also wants you to know that insulin causes $\uparrow$ nuclear transcription, $\uparrow$ serine-threonine kinase activity, and $\downarrow$ ubiquitin-mediated proteolysis.
- Patient has infection + goes into septic shock; Q asks for what LPS binds to that causes shock; answer = Toll-like receptors; in endotoxic shock, the lipid A of lipopolysaccharide (LPS) binds to CD14 on macrophages (toll-like receptor 4; TLR-4), resulting in macrophages releasing cytokines, namely TNF- α and IL-1. Do not confuse endotoxic shock with toxic shock syndrome caused by *S. aureus*, where TSST toxin (superantigen) bridges MHC-II and TCR, causing the macrophage to release cytokines; if the Q mentions cotton nasal packing or tampons, choose the MHC-II + TCR combo, not CD14/TLR-4.
- 2F + missing superior left parathyroid gland; what's the embryologic defect? $\rightarrow$ answer = fourth pharyngeal pouch; fourth pouch = superior parathyroids; third pouch = thymus + inferior parathyroids; easiest student fuck-up = selecting arch instead of pouch; pouches = endoderm; arches = mesoderm; clefts/grooves = ectoderm; DiGeorge syndrome $\rightarrow$ aplasia of 3rd and 4th pharyngeal pouches, leading to hypocalcemia, T cell deficiency with absent thymic shadow, cleft lip/palate, and heart defects (tetralogy of Fallot; truncus arteriosus).
- 20F + recurrent Staph infections + deficiency of CD18; what mechanism best explains this condition? $\rightarrow$ answer = failure of emigration of phagocytes to inflamed tissues; leukocyte adhesion deficiency is caused by deficiency of LFA-1/CD18 integrin, which is required for diapedesis of leukocytes to sites of infection; result is recurrent infections (often Staph) with absent pus; delayed separation of umbilical cord is often omitted from Qs because it's too easy of a detail.
- 40F + develops AML after exposure to agent; what's the most likely agent? $\rightarrow$ answer = benzene. You need to know benzene exposure is associated with AML. Naphthylamine is associated with urothelial cancers (i.e., transitional cell carcinoma); vinyl chloride can cause angiosarcoma of the liver.
- 54F + treated with anti-platelet agent that prevents platelets from interacting with fibrinogen; what's the agent? $\rightarrow$ answer = abciximab $\rightarrow$ monoclonal antibody against glycoproteins IIb/IIIa on platelets; these glycoproteins carry out aggregation and are normally bridged by fibrinogen; abciximab is a fibrinogen analogue; don't confuse aggregation (platelets to each other) via GpIIb/IIIa with adhesion (platelets to endothelium) mediated by Gp Ib. Eptifibatide and tirofiban are lower yield drugs that also inhibit GpIIb/IIIa; abciximab is HY agent with this MOA; HY drugs that inhibit ADP2Y12 receptor on

platelets are clopidogrel and prasugrel; ticagrelor and ticlopidine are lower yield drugs that antagonize ADP2Y12.

- Which of the following is both an anti-platelet agent and vasodilator? → answer = cilostazol; both cilostazol and dipyridamole inhibit platelet phosphodiesterase; cilostazol used for intermittent claudication *after* exercise regimen fails; dipyridamole-thallium is a type of pharmacologic stress test (answer on USMLE if patient needs AAA repair but perioperative MI risk needs to be assessed).
- 57M + asks physician why celecoxib increases risk of adverse cardiovascular events; answer = inhibits platelet prostaglandin synthesis without inhibiting thromboxane synthesis; celecoxib is selective COX2 inhibitor.
- 35F + started taking high-dose vitamins past few months as part of new dietary regimen + has cracked lips, rough/dry skin, and alopecia on scalp; these findings are due to excess of which vitamin? → answer = vitamin A → can cause desquamation, increased hepatic transaminases, dyslipidemia, and pseudotumor cerebri (increased intracranial pressure); teratogenic; must do pregnancy test before commencing isotretinoin for acne.
- 46M + alcoholic + Q shows you hypersegmented neutrophil; what's the vitamin deficiency? → answer = folate (B9); most common vitamin deficiency; B9 deficiency also the answer for history of tea and toast diet for 6 months, and anti-epileptic use (i.e., valproic acid, phenytoin, carbamazepine); B12 is the answer for veganism, strict vegetarianism, chronic gastritis, pernicious anemia, Hx of gastrectomy or ileectomy, Crohn disease, *D. latum* infection.
- 34F + receives phentolamine; what intracellular effect is expected? → answer = ↓ phospholipase C activity; phentolamine is α_1 antagonist; α_1 is coupled to a G-alpha-q G-protein. Agonism will ↑ phospholipase C activity, which will ↑ IP3 and ↑ DAG; therefore antagonism will ↓ phospholipase C activity, ↓ IP3, and ↓ DAG.
- 17F + Hx of heavy menses + easy bruising + ↓ Hb + normal platelets + normal PT + ↑ PTT; what factor is she deficient in? → answer = von Willebrand factor; vWF bridges GpIb on platelets to vascular endothelium; deficiency in vWD causes: ↑ bleeding time, normal PT, and ↑ PTT *about half the time*. Bleeding time is always elevated because the primary role of vWF is to ensure platelet adhesion; PTT is only increased about half the time because vWF's stabilization of factor VIII in plasma is a mere

secondary/ancillary role. Plenty of NBME Qs will have normal PTT in vWD. Platelets can be normal or slightly decreased. Vignette of vWD will always give a mix of one platelet problem (easy bruising, petechiae, epistaxis) and one clotting factor problem (menorrhagia, excessive bleeding with tooth extraction). Hemarthrosis is a clotting factor problem but is seen more in hemophilia rather than vWD. vWD is AD; hemophilia A and B are XR.

- Q asks about which of the following is most likely to disturb Hardy-Weinberg equilibrium; answer = “appreciable rate of gene mutation”; H-W assumes: there is no appreciable rate of gene mutation; mating within a population occurs at random; the population is relatively large; there is no selection against a particular genotype; and there is no significant immigrant population.
- 43M + elevated fasting glucose + arthritis + cardiomegaly; what’s the mechanism for this patient’s condition? → answer = increased intestinal absorption of iron; Dx = hemochromatosis; AR; chromosome 6; Fe can deposit in tail of pancreas, leading to diabetes; hemochromatosis and primary hyperparathyroidism are two HY causes of pseudogout (calcium pyrophosphate deposition disease); this will carry an osteoarthritis-like presentation (i.e., even with findings such as Heberden nodes), but the vignette will be hemochromatosis and primary hyperparathyroidism, so you know the Dx is pseudogout; Fe can deposit in the heart, leading to either dilated or restrictive cardiomyopathy; Fe can deposit in the hypothalamus and/or testes, causing infertility; “Bronze diabetes” = hyperpigmentation of the skin due to hemosiderin deposition → e.g., patient with darkening of the skin of the forearms, increased fasting glucose, and infertility or arthritis; answer = “check serum iron and ferritin.”
- 6M + hyperpigmentation around the lips + colonic polyps; what kind of polyps are they? → answer = hamartomatous; Dx = Peutz-Jeghers syndrome, characterized by perioral melanosis + hamartomatous colonic polyps.
- 26F + hyperpigmentation + BP of 100/60 + muscle weakness; what combination of sodium, potassium, and cortisol are expected in this patient? → answer = ↓ sodium, ↑ potassium, ↓ cortisol; Dx is Addison disease; Na is ↓ and K ↑ because aldosterone is ↓. Condition is often caused by antibodies against 21-hydroxylase, which is needed to synthesize both aldosterone and cortisol in the adrenal cortex.

- 80F + colonic diverticula; which part of the colon most likely has the diverticula? → answer = sigmoid colon.
- Q asks about what would be increased in third trimester of pregnancy; answer = “hepatic protein production”; in contrast, CO₂ is decreased (a mild respiratory alkalosis can be seen in pregnancy); pulmonary residual capacity is decreased because the diaphragm is pushed up; serum osmolality is decreased due to expansion of plasma volume, resulting in dilutional effects; systemic vascular resistance is decreased because the placental vessels add units to the “parallel circuit” of the systemic vasculature.
- 40F undergoing chemotherapy + has low leukocyte count; what drug will help this patient? → answer = pegfilgrastim (granulocyte colony-stimulating factor; G-CSF); will cause an increase in neutrophils.
- 27M + infertility + high serum testosterone + low serum LH and FSH; what’s the most likely cause of this patient’s infertility; answer = exogenous testosterone use; testosterone-secreting tumor could also theoretically be an answer, but only the former listed.
- 25M + has facial nerve and optic nerve on left damaged in car accident; why does he regenerate facial movement over time but not vision? → answer = optic nerve is myelinated by oligodendrocytes; oligodendrocyte-myelinated nerves of CNS have limited regenerative capacity compared to Schwann cell-myelinated nerves of the PNS; facial nerve is myelinated by Schwann cells.
- Drug is given that increases insulin release; what is the MOA of this drug? → answer = increases the ATP/ADP ratio; theoretically, many answers could make sense here; the point is: does the student know the mechanism for insulin release: ATP closes a K⁺ channel on the beta-islet cells → K⁺ builds up within the cell → depolarization → Ca²⁺ then influxes → causes release of insulin-containing vesicles from beta cell; in other words, if a drug increases the ATP/ADP ratio within the beta islet cell, then theoretically that would facilitate insulin release. In real life, sulfonylureas (e.g., glipizide) function to close the ATP-gated K⁺ channel without the need for ATP.
- 46M + esophageal varices; which vein is most likely to have increased blood flow? → answer = left gastric vein; ↑ portal venous pressure causes backup to left gastric vein, which backs up to esophageal veins.
- 74M + lesion on cheek showing neoplastic cells with dense pigment granules; what’s the diagnosis? → answer = melanoma. Sounds like pigmented BCC, but USMLE wants melanoma for this description.

BCC histologically can be described as having islands and nests of basophilic cells; squamous cell carcinoma has keratin pearls and intercellular bridges.

- Woman gets toxoplasmosis but cats isn't listed as an answer; where did she get it from? → answer = pork; can also be written as just simply "delicatessen meats"; both answers on offline NBMEs.
- Biostats Q asks about what increasing the # of participants in a study would result in; answer = decreased standard error of the mean; $SEM = SD / (\text{square root of the \# of samples})$; so if samples increase, then the denominator increases, and SEM decreases. Increasing # of samples will also increase power and decrease beta error (false-negative error).
- 50F + weight loss + progressive shortness of breath + CXR shows pleural-based mass encasing the lungs; what kind of cells are this patient's malignant cells derived from? → answer = mesothelial cells; mesothelioma will classically present with pleural and/or supradiaphragmatic plaques; if shown a gross specimen, mesothelioma will appear to surround the lungs circumferentially, rather than appearing as an isolated mass.
- 30F + last menstrual period 8 weeks ago + USS shows no obvious fetus or gestational sac + curettage shows trophoblastic proliferation; what will chromosomal analysis of this tissue show? → answer = 46XX; Dx = complete mole; empty egg fertilized by one sperm that then duplicates (i.e., genetic material is only of paternal origin); complete mole has *no* fetal parts; partial mole carries 67 chromosomal karyotype → one egg fertilized by two sperm; fetal parts present on USS.
- Pic is shown of cavitating lung cancer; what's the diagnosis? → answer = squamous cell carcinoma; can present with cavitary/cavitating lesion.
- 35F + menorrhagia + deep/hoarse voice + delayed tendon reflexes + large, globular heart with crackles at lung bases + low Hct + low MCV; next best step? → answer = check serum TSH; patient likely has Hashimoto (hypothyroidism); can cause menorrhagia, reduced reflexes, and cardiomyopathy. Anemia in this case is likely IDA due to heavy periods (although anemia of chronic disease does indeed present with low MCV on many NBME Qs); other HY findings for hypothyroidism are hypothyroid myopathy ($\uparrow$ serum CK), $\uparrow$ LDL cholesterol, $\uparrow$ hepatic transaminases, and low mood; findings such as constipation, cold intolerance, brittle hair, and doughy skin are often omitted from Qs because they're too easy.

- Biostats Q where researchers conclude there's no increased risk of cancer in one study group vs another, with a p value of 0.02; what does this p value indicate? → answer = probability of accepting the null hypothesis when it's false. This study concluded that there was no difference between the groups, so they are saying there's a 2% chance they fucked up, where there really was a difference but they just didn't catch it (i.e., beta error, or false-negative error); a p value given for a study conclusion, in general, = the chance your conclusion is wrong; so if you get another biostats Q where they say, yes, there was a difference between two drugs, with p value of 0.01, the interpretation is: there's a 1% chance we fucked up and there really was no difference between these drugs (i.e., a 1% chance we rejected the null hypothesis when it was true).
- Study finds high sugar consumption causes prostate cancer, with p value of 0.07; what does this p value indicate the probability of? → answer = chance of rejecting a true null hypothesis. In other words, a true null hypothesis means there really is no increased risk of prostate cancer caused by sugar. But because the study said there's a difference, they rejected the null hypothesis. And the p value of 0.07 means they're saying there's a 7% chance they acknowledge they fucked up, where they said there was increased risk of cancer even though in truth there isn't.
- Q asks which of the following agents is an NMDA receptor antagonist; answer = ketamine; NMDA receptors are a type of glutamate receptor; ketamine is an NMDA glutamate receptor antagonist.
- Q asks which of the following will occur as a result of giving bortezomib; answer = decreased activation of CD8+ T cells; bortezomib is a proteasome inhibitor that causes decreased MHC I expression.
- 25M + motorcycle accident + now cannot dorsiflex left foot but eversion intact; Q asks which nerve is fucked up; answer = deep peroneal (fibular) nerve; **d**eep does **d**orsiflexion and sensation of the webbing between the first and second toes; superficial peroneal (fibular) nerve does eversion + sensation over most of the dorsum of the foot; common peroneal nerve injury would result in both loss of dorsiflexion and eversion; tibial nerve injury results in loss of plantar flexion and sensation over sole of the foot.
- 21F + vesicular, painful rash across left flank + pic shows you multinucleated giant cells; what's the most likely pathogen this patient should be screened for? → answer = HIV; only answer listed that

makes sense, as VZV not listed; young patient with shingles (herpes zoster; VZV); immunosuppression in young patient resulting in shingles could reflect HIV infection.

- 25M + 3-day Hx of paresthesias of the legs + difficulty swallowing + diminished tendon reflexes + CSF shows protein of 70 mg/dL and cell count of 5/uL; what's the mechanism for this patient's condition?
→ answer = "destruction of myelin and endoneurial lymphocytic infiltrates"; USMLE wants you to know this description of Guillain-Barre; CSF findings are albuminocytologic dissociation (increased protein but normal cells).
- Q asks for which type of receptor agonism is associated with pain and itching; answer = H1 agonism; second generation H1 blockers such as loratadine can be used for anti-allergy effects; first generation agents such as diphenhydramine and chlorpheniramine have strong anti-cholinergic side-effects and can actually be used, for this reason, as treatment for acute dystonia and motion sickness.
- 30M + sensory nerve injury in bar fight; Q asks what's the rate-limiting step in reestablishing normal sensation in this patient; answer = slow anterograde axonal transport; Wallerian degeneration results in breakdown of axonal skeleton and myelin sheath distal to site of nerve injury; nerve regrows anterograde from site of lesion at 1mm/day; slow anterograde transport / regeneration of nerve becomes rate-limiting factor in restoration of function.
- 25M + rash on palms and soles + receives antibiotic Tx with drug he has received in the past; following current Tx, now has fever, tachycardia, and tachypnea; what's the most likely diagnosis? → answer = Jarisch-Herxheimer reaction; answer can also be written as "reaction to lysis of spirochetes"; patient has secondary syphilis (*Treponema pallidum*), as indicated by the palms/soles rash; Tx is with penicillin G; lysis of spirochetes can cause anaphylaxis-like immune response known as J-H reaction.
- 30F + rheumatoid arthritis + currently being managed with infliximab but has not improved; Q asks for which immune mediator is most likely contributing to symptoms; answer = IL-1. Infliximab is a monoclonal antibody against soluble TNF α (not TNF α receptor); even though TNF α plays a role in rheumatoid arthritis, we know TNF α is not the answer here because the patient is already on a med for it; IL-1 and IL-6 are cytokines known to cause inflammation in RA; tocilizumab is a monoclonal antibody against IL-6 receptor that can be used for advanced RA.
 - Normal Tx for RA is two-armed: symptoms and DMARDs (disease-modifying anti-rheumatic drugs).

- For Sx: USMLE wants NSAIDs followed by steroids; these do **not** slow disease progression.
- For slowing disease progression, the first-line DMARD is almost always methotrexate. If this fails or patient has contraindications to it (i.e., pulmonary or hepatic disease), use an anti-TNF α drug.
- Simple Q asking the function of Sertoli cells; answer = produces aromatase.
 - Sertoli cells are stimulated by FSH and produce androgen-binding protein, aromatase, and inhibin. Anti-Mullerian factor is also made by Sertoli cells in fetus.
 - Leydig cells are stimulated by LH and produce androgens.
 - The androgens produced by Leydig cells are converted to estrogens by the aromatase produced by Sertoli cells.
 - Androgen-binding protein binds testosterone locally and keeps concentrations high in the seminiferous tubules, enabling spermatogenesis. Do not confuse ABP with sex-hormone-binding globulin, which is produced by the liver and binds androgens and estrogens for transport around the blood.
 - Leydig cells in males = theca interna cells in females.
 - Sertoli cells in males = granulosa cells in females.
 - Sertoli-Leydig cell tumor (seen in females as well, despite the name) produces androgens $\rightarrow$ leads to hirsutism/virilization; can also cause gynecomastia because $\uparrow$ androgens can be aromatized to estrogens.
 - Granulosa cell tumors classically present on USMLE as endometrial hyperplasia due to unopposed / higher estrogen levels $\rightarrow$ can cause endometrial cancer (bleeding per vaginum).
- 20M + both common carotids are compressed externally for 10 seconds; what effect will this have on alveolar ventilation, venous tone, arterial tone, and blood pressure; answer = all are $\uparrow$.
 - If blood flow is $\downarrow$ to the carotid sinus (area just superior to bifurcation of common carotid bilaterally) $\rightarrow$ $\downarrow$ stretch of baroreceptors $\rightarrow$ $\downarrow$ CN IX (glossopharyngeal) afferent firing to solitary nucleus of medulla (**not** $\uparrow$ firing, because the rate of firing is stretch-dependent) $\rightarrow$ $\downarrow$ efferent CN X (vagus; parasympathetic) + $\uparrow$ efferent sympathetic firing to cardiac nodal tissue $\rightarrow$ $\uparrow$ heart rate.

- ↑ sympathetic efferent firing also causes ↑ arteriolar and venous vasoconstriction via α_1 agonism ($\uparrow$ SVR $\rightarrow$ $\uparrow$ BP) + ↑ bronchodilation and respiratory rate via β_2 agonism.
- 39M + bilateral renal enlargement + high blood pressure + father died from subarachnoid hemorrhage; this patient most likely has a gene mutation in which protein? $\rightarrow$ answer = polycystin; diagnosis is ADPKD; high blood pressure is from cysts impinging on the renal microvasculature leading to surges in RAAS.
- 52M + 3-month-Hx of burning in esophagus + Hx of diabetes + ↓ pin-prick discrimination up to ankles + HbA1c of 9.6%; what's the MOA of the most appropriate pharmacologic therapy? $\rightarrow$ answer = antagonism of D2 receptors; drug is metoclopramide; Dx is diabetic gastroparesis; metoclopramide is both a pro-kinetic and anti-emetic.
 - In contrast, if vignette is simply 42M + GERD-like presentation + no mention of poor diabetic control, answer is "trial of PPIs" or "trial of H2 blockers" (I've seen both as correct answers, but if you're forced to choose between the two, always choose PPIs) for GERD.
- 30F + cuts finger on paper; moments after the cut, the bleeding stops; which of the following is the mechanism for the quick cessation of blood loss? $\rightarrow$ answer = localized secretion of endothelin; sounds slightly unusual, as endothelin is notably increased in pulmonary vascular resistance, but it acts peripherally as well.
 - Endothelin is ↑ in pulmonary hypertension, cor pulmonale, and congestive heart failure (including isolated left heart failure); essentially any cause of increased pulmonary pressure leads to ↑ compensatory endothelin release (to restrict the increased blood flow).
- 28M + renal failure + hematuria + hearing loss; what's the mechanism for this patient's disease? $\rightarrow$ answer = mutation in type IV collagen (Alport syndrome, X-linked); wrong answer is antibodies against type IV collagen (Goodpasture); Alport $\rightarrow$ hematuria in a male + ear and/or eye problem; Goodpasture $\rightarrow$ male 20s-40s with hematuria and hemoptysis.
- 19F + alkaline urine + renal tract calculi; which organism is most likely responsible? $\rightarrow$ answer = Proteus; other correct answers could be Klebsiella or Serratia; struvite (ammonium magnesium phosphate) stones form in alkaline urine secondary to UTIs caused by urease-positive organisms.

- 62F + high Hb + high WBCs + high platelets + pO₂ 93%; what combination of EPO and arterial O₂ blood content is most likely in this patient? → answer = ↓ EPO + normal arterial O₂ content; diagnosis is polycythemia vera (JAK2 mutation), which is a bone marrow overproduction of the different cell lines; the high primary RBC production by the bone marrow suppresses EPO production by the kidney; WBCs and/or platelets will also be increased. In contrast, in secondary polycythemia caused by lung disease (↑ EPO due to ↓ O₂) or renal cell carcinoma (↑ EPO production by cancer), only RBCs are high. High hemoglobin (>17.5 g/dL) is seen + Hb-O₂ saturation that is at the lower end of normal (93% is minimally acceptable); arterial O₂ content = pO₂ (dissolved O₂) + Hb-O₂ (amount bound to Hb); since Hb is high, less is saturated, but the amount of O₂ in the blood (content) is still the same because there is no lung disease, and Hb can still bind O₂ without a problem. Tx for polycythemia vera is serial phlebotomy.
- What is the role of IκB in relation to NFκB? → answer = “releases NFκB after undergoing phosphorylation”; USMLE also wants you to know that corticosteroids ↓ NFκB signaling in leukocytes.
- Newborn + chorioretinitis + cataracts + microcephaly + no other info; Dx in the neonate? → answer = CMV; congenital Toxo always = triad of chorioretinitis + hydrocephalus + intracranial calcifications; so if you do not get this exact latter triad, you can eliminate congenital Toxo. Congenital CMV tends to have a mix of non-specific findings, e.g., deafness, chorioretinitis, intracranial calcifications. Congenital rubella will have PDA; congenital syphilis will have tooth abnormalities (Hutchinson), and often saddle nose and/or Saber shins. Congenital VZV will have zig-zag skin lesions, limb abnormalities, and microphthalmia.
- 14M + temp 103F + WBCs 25,000 with neutrophil shift; what’s the mechanism for the high leukocytes in this patient’s blood? → answer = accelerated release of leukocytes from bone marrow postmitotic reserve pool; diagnosis is leukemoid reaction, which means leukocytosis in the setting of infection; leukocyte ALP is high; smear can show neutrophilia (i.e., a smear showing just, e.g., 8 neutrophils).
- Q asks what the anticodon sequence of tRNA does; answer = selection of specific mRNA for ribosomal binding; tRNA anticodon is a 5′→3′ sequence that is complimentary to the codons of mRNA; when a codon of an mRNA codes for a specific amino acid, this is because a corresponding (and complimentary) tRNA binds to it via its anticodon site. The tRNA has an amino acid bound at its

acceptor stem. The tRNAs then transfer amino acids to the growing peptide strand at the ribosomes during protein synthesis.

- 15M + skiing accident + tachycardia + tachypnea + BP 98/72 + decreased breath sounds on right + CXR shows decreased lung markings on one side and large, black pleural space; Dx? → answer = tension pneumothorax; blood pressure can be low or low-normal; tension pneumothorax is a pneumothorax in which hemodynamic decompensation occurs (i.e., low BP); in this patient, although BP is technically low-normal, this is only because the HR is increased to compensate, and the patient will likely continue to decompensate. Mechanism for low BP in tension pneumothorax is compression of venous structures returning blood to right atrium; Q need not mention contralateral tracheal deviation; breath sounds will be decreased on affected side because air does not move in and out through the pleural space; Tx is needle decompression followed by chest tube connected to underwater apparatus.
- 36M + stab wound to chest + low BP + JVD + systolic BP drops 12 mmHg on inspiration; Dx? → answer = cardiac tamponade; Beck triad for cardiac tamponade = hypotension, JVD, muffled heart sounds; Q need not mention all three. Pulsus paradoxus, with drop of systolic BP >10 mmHg with inspiration is classically seen in tamponade; tamponade is simply a pericardial effusion + hemodynamic decompensation (i.e., low BP, or low-normal BP with high HR to compensate); ECG shows electrical alternans (alternating QRS amplitudes due to blood moving within the pericardial sac, causing fluctuation in distance between the leads); ECG will show same finding in simple pericardial effusion that is not tamponade; Tx is echo to confirm followed by pericardiocentesis or pericardial window. Factor that is most important for tamponade development is acuteness of blood accumulation, not volume – i.e., stab wound, or LV free-wall rupture post-MI, can lead to small volume of blood accumulating quickly in pericardial sac, leading to tamponade, whereas, e.g., lymphatic obstruction from lymphoma could cause slow-accumulating, high-volume chylous effusion over many months that does not cause tamponade.
 - Hemothorax is contrasted with tamponade in that in the former, there are flattened neck veins, whereas in tamponade, there's JVD. This is how to distinguish if they don't mention the CXR or decreased unilateral breath sounds for hemothorax.

- 42F + in hospital for surgery two days ago + shortness of breath + HR 92 + ECG shows no abnormalities; the medication she needs has what MOA? → answer = activation of anti-thrombin; heparin is drug given; diagnosis is pulmonary embolism; sinus tachycardia is most common finding on ECG (normal ECG + ↑ HR = sinus tachycardia); after heparin, do spiral CT of chest for diagnosis; do V/Q scan first in pregnant women.
- 29M + lancinating/stabbing pain in cheek lasting several seconds + happens spontaneously; Q asks location of problem; answer = pons; unusual answer, since trigeminal neuralgia is a peripheral mechanism, but CN V comes from pons; answer can also be “microvascular compression of CN V”; no acute Tx since the Sx subside so quickly; prophylaxis is carbamazepine.
 - Do not confuse trigeminal neuralgia of the V1 distribution with cluster headache; the latter will occur usually during sleep and will last up to half hour; details such as lacrimation and rhinorrhea are HY but too easy and often omitted from Qs; Q can mention pupillary changes during cluster headache; Tx = 100% oxygen; prophylaxis is verapamil (not propranolol).
 - Migraine prophylaxis on USMLE = propranolol (sometimes just written as beta blockade); Tx is NSAIDs followed by triptans (serotonin receptor agonists).
- 23F + relaxing after coming back from a run; which of the following describes her changes in diastole and systole as her heart rate returns to normal? → answer = diastole increases more than systole; the Q can also be asked as the inverse: that is, as HR increases, diastole shortens more than systole; the learning point is that the slower the heart rate, the greater the proportion of time spent in diastole.
- 40F + goes for run; which of the following regarding diastole and systole represent her coronary artery filling pressure at rest vs while running? → answer = at rest: diastole > systole; while running: diastole > systole; answer is the same for both; the coronary arteries fill during diastole irrespective of how fast the HR is.
- 50M + severe upper back pain + 3/6 holo-diastolic murmur loudest after S2; Dx? → answer = aortic dissection; classically pain radiating back between the scapulae; retrograde propagation of the dissection can cause aortic root dilation with aortic regurgitation (decrecendo holo-diastolic murmur); CXR shows widening of the mediastinum; give labetalol to decrease shearing forces; increased risk in connective tissue disorders such as Marfan and Ehlers-Danlos; greatest risk factors are hypertension and cocaine use.

- 19M + comes to physician with chest pain after night of heavy partying + walks through the door bent over at the waist; next best step in diagnosis? → answer = ECG showing diffuse ST-elevations; diagnosis is pericarditis; pain is lessened when leaning forward + worsened when leaning back; ST-elevations will be seen in most leads, as opposed to 3-4 contiguous leads, as with MI; cocaine is risk factor for serous pericarditis; Tx with NSAIDs, corticosteroids, and colchicine.
- 37F + rheumatoid arthritis + PR-depressions and ST-elevations seen in many leads on ECG; diagnosis? → answer = pericarditis; common in autoimmune diseases such as RA; PR-depressions are highly specific for pericarditis but not common; diffuse ST-elevations seen in all cases on USMLE if they mention ECG findings.
- 34F + severe chest pain + worse when reaching behind her back and above her head; diagnosis? → answer = costochondritis → MSK because changes with position (pericarditis only exception when cardiac).
- 34F + chest pain that does not change with position + chest pain worse with palpation; diagnosis? → answer = costochondritis; even though it doesn't change with position, it worsens with palpation.
- 53M + recent surgery + in hospital; Q asks for what post-surgical finding is most likely to reflect good recovery; answer = brisk diuresis; USMLE wants you to know that diuresis (urination) is a good sign of post-operative recovery. This is also the case in patients who have renal insufficiency who are being managed – i.e., brisk diuresis indicates recovery from the renal pathology.
- 40M + history of alcoholism + serum potassium of 3.2; following K⁺ administration, potassium remains at 3.2; why? → answer = hypomagnesemia; low Ca²⁺ and K⁺ non-responsive to supplementation → answer = check serum magnesium; Mg²⁺ required for basal level PTH secretion (↓ Mg → ↓ Ca); Mg²⁺ also inhibits a K⁺ secretory pump in the distal kidney (↓ Mg → ↑ K wasting); alcoholics are classic group at risk of hypomagnesemia due to mere dietary deficiency (EtOH is 7kcal/g, so they fill up on alcohol).
- 18M + gets lost hiking in woods for 3 weeks + BMI 27; what electrolyte are we worried about upon restoration of oral intake? → answer = hypophosphatemia; refeeding syndrome can cause various electrolytes to be low, but phosphate is the main one; mechanism is rapid sequestration of phosphate by glycolytic intermediates as glycolysis resumes.

- 31F + Hx of ITP + has splenectomy + platelet count goes back to normal; 6 months later, she has low platelets again; why? → answer = accessory spleen; 10% of people have a second, accessory spleen that is less than a centimeter in size; after splenectomy, the accessory spleen can grow in size; if the USMLE Q gives you ITP that was definitively managed with splenectomy, but then the ITP “comes back,” the answer is accessory spleen.
- 49M + long history of alcoholism + presents with smelly stools + pancreatic enzymes are normal; Dx? → answer = chronic pancreatitis → exocrine pancreas insufficiency due to repeated bouts of acute pancreatitis; pancreatic enzymes low or low-normal; steatorrhea from lack of lipases in pancreatic secretions; Tx with pancrelipase.
- 49M + alcoholism + had pancreatitis a week ago and recovered well with bedrest; now has fever + leukocytosis + increased serum lipase and amylase; diagnosis? → answer = pseudocyst; this is a common complication of acute pancreatitis; called a pseudocyst because it doesn’t have actual walls; the pancreatic parenchyma encases the fluid collection; can be drained internally via ERCP (2CK surg NBME).
- 40F + found in street with hypothermia + rewarmed in hospital + picture is shown of her feet and they’re red; what electrolyte are we most worried about? → answer = hyperkalemia → reperfusion injury causes oxygen radicals that increase the risk of rhabdo, especially in alcoholics.
- 82F + found in her house by the staircase; urine blood is 2+; RBCs on urine light microscopy are 3-4 per high-power field; diagnosis? → answer= rhabdomyolysis; classic finding is false-positive blood on urine dipstick; USMLE can give 3-4 RBCs/hpf as false-positive.
- 44M + recent surgery in hospital + tachycardia + fever + high BP; what drug should be administered? → answer = chlorthalidone; Dx is delirium tremens (alcohol withdrawal); Tx is benzo (diazepam, lorazepam, chlorthalidone). If the Q asks why tremulousness (tremor) present in delirium tremens, the answer is beta-2 agonism.
- 13F + asthma attack + tremor + albuterol + IV steroids given for acute asthma attack; why is there tremor? → answer = albuterol; tremor is most common side-effect of beta-2 agonists like albuterol.
- 57F + on various medications + comatose + seizure + prolonged QRS complexes on ECG; what med is responsible? → answer = amitriptyline; TCAs cause CCC → coma, convulsions, cardiotoxicity; ECG findings can be manifold; Tx with sodium bicarb (decreases binding of TCA to myocardial sodium

channels; increased urinary excretion via alkalinization is wrong fucking answer and refers to NSAIDs;

TCA's are basic, not acidic like salicylates).

- 22M + on anti-depressant med + has suprapubic mass + face is hot and red; which drug is he on? → answer = amitriptyline (any TCA is answer); TCA's also have nasty anti-cholinergic side-effects → *opposite of DUMBBELSS* (refers to *cholinergic* side-effects) → diarrhea, urination, myosis, bradycardia, bronchoconstriction, excitation of skeletal muscle, lacrimation, salivation, sweating; so TCA's can cause constipation, urinary retention, mydriasis, tachycardia, bronchodilation (in theory, but this is more sympathetic beta-2-mediated, rather than anti-muscarinic), relaxation of skeletal muscle tone, dry eyes, dry mouth, anhidrosis (leading to heat retention and red, hot patient).
- 30F + going on a plane + receives diphenhydramine to prevent motion sickness; what receptor is responsible for anti-motion sickness in this patient? → answer = anti-muscarinic; diphenhydramine and chlorpheniramine are first-generation H1-blockers with nasty anti-cholinergic side-effects; muscarinic receptor blockade ameliorates feelings of motion sickness; H1 will not be listed as an answer in this case, so choose muscarinic; H2 might be listed, but diphenhydramine and chlorpheniramine don't act on H2.
- 24M + cuts his arm while scraping the BBQ in the backyard; which cell infiltrating into the area of the wound in the acute setting will most likely decrease his risk of a bacterial infection? → answer = neutrophils; present in acute inflammation; part of innate immunity; phagocytose and kill bacteria via lysosomal hydrolases and H₂O₂ production.
- 55M + hyperthyroidism + pulses absent in left lower extremity; Dx? → answer = arterial embolic occlusion; acute limb ischemia in this case is caused by left atrial mural thrombus, secondary to hyperthyroidism-induced atrial fibrillation, launching off to the femoral artery; absence of pulses indicates arterial occlusion; in contrast, in DVT, pulses are **normal**, since the arteries are fine; AF can also lead to clot launching off to brain (stroke) or SMA/IMA (acute mesenteric ischemia).
- 80M + atrial fibrillation + severe abdominal pain + physical exam shows mild tenderness to palpation; Dx? → answer = acute mesenteric ischemia; pain out of proportion to physical exam is characteristic; left atrial mural thrombus from AF launched off to SMA or IMA.
- 65M + diabetes + Hx of CABG + intermittent claudication + pain in abdomen 1-2 hours after meals; most likely cause of abdominal pain? → answer = atherosclerosis; Dx is chronic mesenteric ischemia;

presentation will sound like duodenal ulcer (pain 1-2 hours after meals) but in patient who has severe cardiovascular Hx. *H. pylori* causing duodenal ulcer, in contrast, will be a younger immigrant patient usually.

- 30M + rheumatoid arthritis + overdosed on methotrexate in suicide attempt + now has massive hepatonecrosis; the liver can regenerate via which of the following cellular mechanisms? → answer = recruitment of cells from G0 into cell cycle; when there is hepatocellular damage, remaining cells in G0 (quiescent phase) will enter G1 of the cell cycle and begin replication; the wrong answer is stem cells entering G1 of cell cycle; this is because stem cells aren't activated into G1; it is merely the case that other hepatocytes exist in G0, not undergoing replication, and are then stimulated into G1 for replication.
- 500 factory workers are surveyed to determine whether they smoke or not and whether they have COPD or not; what kind of study design is this? → answer = cross-sectional survey; this is merely a snapshot of a population at one point in time; a survey group is called up, and then numbers of smokers vs non-smokers, and disease vs no disease, can be assessed. This determines prevalence of disease but not causation. This is different from case-control, where the initial study design is not that of calling up people and asking all of this info at once; instead, case-control entails taking two distinct groups – one with COPD vs one without COPD, and then asking about their smoking history to attempt demonstrate causation.
- 20F + insulinoma; Q asks about C-peptide, insulin, and serum ketone levels; answer = ↑ C-peptide, ↑ insulin, ↓ ketones. C-peptide is a cleavage protein during the production of endogenous insulin, so if endogenous insulin is produced (i.e., via insulinoma), then C-peptide should be proportionally high. Ketogenesis is inhibited by insulin, so ketones must be low. Ketone production requires fatty acid breakdown into acetyl-CoAs as the starting blocks; beta-oxidation is inhibited by malonyl-CoA, which is an intermediate in fatty acid synthesis, as stimulated by insulin.
 - A tangential HY point is that DKA is not seen in type II diabetes because insulin is high initially; DKA is only seen in type I, since insulin is deficient. If a USMLE Q gives you acute type II diabetes and asks what's high in the patient's serum, the wrong answer is ketones; the correct answer is insulin (hyperinsulinemia due to insulin resistance).

- 2M + anemia + erythroblasts containing ferritin granules + enzyme analysis shows decreased delta-aminolevulinic acid synthase; administration of which vitamin might help this patient's condition? → answer = B6; diagnosis is X-linked sideroblastic anemia; first step of heme synthesis is succinyl-CoA + glycine, via delta-ALA synthase and vitamin B6, becomes delta-aminolevulinic acid; sideroblastic anemia is due to failure to synthesize delta-ALA and can be X-linked or classically caused by alcohol. Prussian blue staining of ringed sideroblasts is classic → iron cannot be incorporated into heme and remains within the RBC precursor around the mitochondria in a ring-distribution.
 - USMLE likes the heme synthesis disorders:
 - Acute intermittent porphyria → autosomal dominant; deficiency of uroporphobilinogen deaminase; buildup of uroporphobilinogen; Sx are abdominal pain, red urine, neuropathy. Tx w/ glucose and hematin can help.
 - Porphyria cutanea tarda → autosomal dominant; deficiency of uroporphobilinogen decarboxylase; buildup of uroporphyrins and coproporphyrins; Sx are photosensitivity and tea-colored urine.
- 62F + flank mass + serum calcium of 11.5 mg/dL + hemoglobin of 18.5 g/dL; diagnosis? → answer = paraneoplastic syndrome; renal cell carcinoma can produce PTHrP (correct, similar to SCC of the lung) and EPO (causing high hemoglobin). USMLE might give you a vignette of a smoker who has hypercalcemia and polycythemia + show you a histologic specimen of many clear cells; clear cell carcinoma is the most common variant of RCC; USMLE likes you to know histo for RCC is clear cells. RCC can classically present with flank pain. Wilms tumor in a kid is painless flank mass.
- Neonate born at 28 weeks gestation + difficulty breathing; what is deficient in this patient? → answer = lamellar bodies; lamellar bodies are specialized organelles within type II pneumocytes that produce surfactant; dipalmitoylphosphatidylcholine (lecithin) / sphingomyelin ratio should be >2.0-2.4 for NRDS probability to be substantially decreased; elastic recoil is *increased* in NRDS due to lack of surfactant; surfactant normally prevents the hydrostatic forces of the alveoli from causing collapse.
- 30F + pancreatitis + ARDS; following complete resolution of the ARDS, what is most likely to be increased in this patient's lungs? → answer = surfactant protein D (marker of lung injury); weird, but know it because it's on Step 1 NBME.



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MEHLMANMEDICAL HY USMLE REVIEW PART IV

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